

EC5.206 – Introduction to Statistical Signal Processing – Final Exam

Date: 2nd May, 2026

Instructor: Santosh Nannuru

Maximum marks: 50

Exam duration: 180 minutes

Instructions:

- Mention any additional assumptions you make.
 - Clearly show the steps used to arrive at your solution.
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1. [12 Marks] Consider the random process given by

$$x[n] = v[n] \cos(\omega_0 n) + w[n],$$

where $v[n]$ and $w[n]$ are zero-mean random processes and are independent of each other. Here, ω_0 is a known frequency. It is given that $v[n] = a, \forall n$ where a is uniform in $[0, \theta]$, and $w[n] = b, \forall n$ where b is Gaussian with mean 0 and variance 1.

- [2] Sketch two realizations of this random process.
- [3] If the processes $v[n]$ and $w[n]$ have the same autocorrelation function, find θ .
- [5] Find the mean and autocorrelation function for the random process $x[n]$.
- [2] Comment on the wide sense stationarity of this process. How does it depend on the value of $\omega_0 \in [0, 2\pi)$?

2. [12 marks] A stationary ARMA process $x[n]$ follows the difference equation

$$x[n] = x[n-1] + ax[n-2] + v[n] + bv[n-1]$$

where $v[n]$ is the zero-mean white noise with unit power and a and b are unknown constants. Assume that the auto-correlation $r_x[k]$ and cross-correlation $r_{vx}[k]$ are known.

- [6] Derive 4 linear equations in the variables a and b . You don't have to solve them.
- [3] Which two of the above 4 equations would you use to solve for a and b such that the total number of auto-correlations and cross-correlations required is minimum.
- [3] If it is known that $r_x[k] = (\frac{1}{2})^{|k|}$ and $r_{vx}[k] = \frac{1}{2}\delta[k] + \frac{1}{3}\delta[k-1] + \frac{1}{3}\delta[k+1]$, solve for a and b using the equations identified in part (b).

3. [12 marks] An estimate of the process $d[n]$ is to be obtained by linear filtering of the noisy measurements $x[n] = d[n] + v[n]$ where $v[n]$ is the noise (uncorrelated with the desired signal). Assume that all the signals are stationary. An *anti-causal* FIR filter is to be designed for this purpose. The filter output $\hat{d}[n]$ is related to the input as follows

$$\hat{d}[n] = \sum_{k=-p}^0 h[k]x[n-k]$$

where $h[k]$ is the p th order anti-causal FIR filter.

- (a) [6] Derive the Wiener-Hopf equations for the filter that minimizes the mean square error given by $J = \mathbf{E}\{|e[n]|^2\} = \mathbf{E}\{|d[n] - \hat{d}[n]|^2\}$.
- (b) [2] How do these equations compare with the *causal* FIR Wiener filter of same length?
- (c) [4] For $p = 2$, $r_d[k] = (0.2)^{|k|}$, and $r_v[k] = \delta[k]$, write down the equations to be solved.

4. [14 Marks] Consider the design of a least squares FIR filter with input signal $x[n]$ and desired signal $d[n]$. The signals $x[n]$ and $d[n]$ are real valued. The least squares filter is determined based on the following modified cost function,

$$J(n) = \sum_{i=0}^n e^2[i] + \alpha \underline{\mathbf{h}}_n^T \underline{\mathbf{h}}_n$$

where $e[i] = d[i] - \underline{\mathbf{h}}_n^T \underline{\mathbf{x}}_n$, $\underline{\mathbf{h}}_n = [h_n[0], h_n[1], \dots, h_n[p-1]]^T$ is the filter coefficient vector at time n , $\underline{\mathbf{x}}(n) = [x[n], x[n-1], \dots, x[n-(p-1)]]^T$ is the input signal vector, and p is the order of the FIR filter. The parameter $\alpha > 0$ is called the regularization parameter. Answer the following:

- (a) [8] Derive the optimal least squares filter $\underline{\mathbf{h}}_n$ that minimizes $J(n)$ for each n (express your answer in matrix form).
- (b) [2] How does value of α affect the filter behaviour? Analyse the cases (a) $\alpha \rightarrow 0$ and (b) $\alpha \gg 1$ to solve for the filters.
- (c) [4] Develop a recursive version of this filter.